PRODUCT REQUIREMENT DOCUMENT

PRD — AI-Powered Marketing Compliance Platform

PRD-2026 · AIAvenue Agency

Executive Summary

Insurance Company requires a marketing compliance platform that automates the review of all outward-facing marketing assets (print ads, social media creatives, banners, brochures, videos, emailers) against 12 distinct IRDAI, brand, and legal rule categories. Currently, manual compliance review takes 3–5 business days per asset, creates bottleneck friction between the marketing and compliance teams, and exposes the company to regulatory penalties from IRDAI. This platform replaces manual workflows with an AI-powered pipeline that scans assets via OCR (Tesseract) and computer vision (OpenCV), surfaces violations with bounding-box annotations, enforces a Maker–Checker–Admin–Director role-based approval hierarchy, streams scan progress via Server-Sent Events (SSE), maintains an immutable audit trail, and sends email notifications via Resend. The stack is Next.js 16 (frontend), Python FastAPI (backend ML inference), AWS S3 (asset storage), and Upstash Redis (scan queue and real-time state). Target TAM is ₹120 Cr, SAM ₹18 Cr, SOM ₹4.5 Cr in the Indian insurance compliance-software market.

Problem Statement

Core Problem

Life insurance marketing in India is one of the most heavily regulated advertising domains. Every piece of marketing collateral — before going live — must be checked against:

- **IRDAI (Insurance Regulatory and Development Authority of India) guidelines** — 8 specific circulars covering mis-selling, benefit exaggeration, mandatory disclaimers, comparison norms, and prohibited language.
- **Brand guidelines** — Logo placement, colour palette, typography, tone of voice, tagline usage.
- **Legal disclaimers** — Statutory warning placement, font-size compliance, risk factor disclosure.

User Pain Points

Persona	Pain Point	Frequency	Severity
Marketing Manager	3–5 day approval cycle delays campaign launches	Per asset	High
Compliance Officer	Manually reviews 40+ assets/week; eye fatigue causes misses	Daily	Critical
Agency Partner	No visibility into why an asset was rejected; resubmission guesswork	Per rejection	Medium
CMO/Director	Cannot audit historical compliance decisions without digging through email chains	Quarterly	High
Legal Team	IRDAI circular updates require manual rule-book updates; no version control	Semi-annual	Medium

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User Pain Points

Impact of Status Quo

- Average compliance cycle: **3.8 days** per asset
- Missed violations reaching market: **estimated 1 in 12 assets** (internal audit data)
- IRDAI penalty exposure: **₹5 Lakh–₹25 Lakh per violation** (Section 45 of Insurance Act 1938)
- Marketing opportunity cost: **₹2.1 Cr/year** in delayed campaign launches

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User Research & Personas

Persona 1: Ananya Sharma — Marketing Manager

Attribute	Detail
Age	31
Role	Senior Marketing Manager, Insurance Company
Tech Comfort	High; uses Monday.com, Canva, Meta Ads Manager daily
Workflow	Creates 8–12 assets/week with agency partners; submits via email for compliance review
Frustration	"I lose 2 days every week just chasing compliance stamps. By the time I get approval, the campaign window is half gone."
Need	Real-time visibility into scan progress, clear violation descriptions, one-click resubmit

Persona 1: Ananya Sharma — Marketing Manager

Persona 2: Rajesh Iyer — Compliance Officer

Persona 3: Vikram Mehta — CMO

Market Analysis

Total Addressable Market (TAM)

The Indian RegTech (Regulatory Technology) market for the BFSI sector is estimated at **₹1,200 Cr** (2026), growing at 22% CAGR. Marketing compliance within life insurance represents an estimated **10%** of this segment.

TAM = ₹120 Cr

Serviceable Addressable Market (SAM)

Insurance Company competes in the private life insurance segment (24 private players). Assuming **15%** of these players are at a digital maturity level ready to adopt AI compliance tools:

- 24 players × 15% = 3.6 ≈ 4 players
- Average annual compliance-software spend per insurer: **₹4.5 Cr**
- SAM = 4 × ₹4.5 Cr = **₹18 Cr**

Serviceable Obtainable Market (SOM)

With a first-mover advantage in AI-powered marketing compliance (most competitors offer document-scanning, not creative-asset scanning):

- Realistic 5-year penetration of SAM: **25%**
- SOM = ₹18 Cr × 25% = **₹4.5 Cr**

Top-Down Validation

Layer	Calculation	Value

Total Indian BFSI RegTech spend	Industry reports (2026)	₹1,200 Cr
Marketing compliance sub-segment	10% of BFSI RegTech	₹120 Cr
Life insurance share	35% of insurance marketing	₹42 Cr
Digitally-ready insurers	15% of 24 players	~4 players
Our obtainable share	25% of SAM over 5 years	₹4.5 Cr

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Top-Down Validation

Bottom-Up Validation

Market Rationale

- IRDAI has increased penalty frequency by **40%** in FY2025–26
- Insurance marketing spend in India crossed **₹3,500 Cr** in 2025; compliance cost as a % of marketing is rising
- No existing off-the-shelf product solves marketing-asset-specific compliance in the Indian insurance context
- InsurTech investment in India hit ₹1,050 Cr in 2025; compliance tools are the next logical wave
- The compliance software market in BFSI is fragmented across document management (DMS), leaving creative-asset compliance as an open white space

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Competitive Analysis

Competitor	Type	Strengths	Weaknesses	Our Edge
DocFox	General compliance automation	Strong KYC/AML; established in banking	No marketing-asset scanning; no IRDAI rule engine; no India-specific	IRDAI-specific rule library + vision-based asset scanning
ComplianceWave	Document management	Good audit trail; enterprise SSO	Rule engine is manual (human-tagged); no AI/OCR	Fully automated AI scanning with bounding-box violations
Zoho Legal	General legal software	Low cost; widespread India presence	Generic contract review; no insurance-specific rules; no image analysis	Image + text analysis with insurance-domain rules
IRDAI Portal (manual)	Regulatory portal	Free; official	No automation; no tracking; no analytics	End-to-end workflow automation with real-time updates
HighRadius	Finance automation	Strong AI; large enterprise base	Focused on order-to-cash; not marketing compliance	Niche focus on marketing compliance for insurance

Detailed Competitive Positioning Map

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HIGH COMPLIANCE DEPTH ▲ | | ◎ Us (AI Compliance Platform) | DOCUMENT FOCUS ────┼──── CREATIVE ASSET FOCUS | | ◎ DocFox | ◎ ComplianceWave | | ◎ Zoho Legal | ◎ IRDAI Portal | ▼ LOW AUTOMATION
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Our Competitive Moat

Key Differentiators:

- 12-category rule engine (8 IRDAI + 2 brand + 2 legal)
- OpenCV bounding-box violation overlay on images
- SSE real-time scan progress (no polling)

- Immutable audit trail (append-only database design)
- Role-based workflow (Maker → Checker → Admin → Director)
- Upstash Redis for scan-queue state management

Solution Overview

High-Level Architecture

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│ Frontend (Next.js 16) │
│ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
│ Asset │
Violation│ Dashboard│ Role-Based UI │ Upload │ Viewer │ & Audit │ (Maker/Checker/ │ (S3 Pre- │ (Bounding│ Log │ Admin/Director) │ signed) │ Boxes) │
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API Gateway
(FastAPI / Next.js API) │
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│ Upstash │ AWS S3 │ Python FastAPI │ Redis │ (Assets) │ (ML Inference) │ (Queue + │ SSE State) │
└──────────┘ └──────────┘ OpenCV + Tesseract OCR Engine └──────────┘ 12-Rule Engine └──────────┘
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Key Features

Feature	Description
Asset Upload	Drag-and-drop upload of PNG, JPG, PDF, MP4; pre-signed S3 URLs for direct upload
AI Scan Engine	Tesseract OCR extracts text; OpenCV detects images, logos, layouts; 12-rule engine evaluates each
Real-time SSE	Server-Sent Events stream scan phases (Uploading → OCR → Rule Check → Report); 100ms granularity
Violation Viewer	Bounding-box overlay on original asset; colour-coded by severity (Red=Critical, Amber=Warning, Blue=Info)
Role Workflow	Maker submits → Checker reviews → Admin approves → Director overrides (optional); email notifications at each step via Resend
Immutable Audit	Append-only log with timestamp, userId, action, before/after state; tamper-evident via SHA-256 chain
Rule Engine Admin	Compliance team can add/edit rules via UI; versioned; activation/deactivation toggle

Dashboard	Rejection rate, average cycle time, violation heatmap by rule category, trend charts
Notifications	Resend-powered email alerts for status changes, escalations, pending approvals

High-Level Architecture

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(FastAPI / Next.js API) │ │ │ └──────┬──────┴──────┴──────┴──────┘ │ └──────┴──────┴──────┴──────┴──────┘ │ │ │ ┌──────▼──┐
┌──▼──────┐ ┌──────▼──────┐ │ Upstash │ │ AWS S3 │ │ Python FastAPI │ │ Redis │ │ (Assets) │ │ (ML Inference) │ │ (Queue + │ │ │ │ │ │ SSE State) │ │ │ │ ┌──────────┐ │
└─────────┘ └─────────┘ │ │ OpenCV + │ │ │ │ Tesseract │ │ │ │ OCR Engine │ │ │ └──────────┘ │ │ ┌──────────┐ │ │ │ 12-Rule │ │ │ │ Engine │ │ │ └──────────┘ │
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Key Features

Feature Deep-Dive — RICE Priority Matrix

Feature	Reach (per month)	Impact (1–5)	Confidence (%)	Effort (weeks)	RICE Score
AI Scan Engine (OCR + 12 Rules)	120 assets	5.0	90%	6	90.0
Violation Viewer (Bounding Box)	120 assets	4.5	85%	3	153.0
Role-Based Workflow (Maker/Checker/Admin/Director)	35 users	4.0	95%	4	33.3
SSE Real-Time Progress	120 assets	3.5	80%	2	168.0
Resend Email Notifications	35 users	3.0	90%	1	94.5
Immutable Audit Trail	5 auditors	3.5	95%	3	5.5
Rule Engine Admin UI	5 users	4.0	70%	4	3.5
Dashboard & Analytics	15 users	3.5	80%	3	14.0
S3 Pre-Signed Upload	120 assets	2.5	95%	1	285.0
Batch Scanning	120 assets	4.0	60%	5	57.6

RICE Formula

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`RICE Score = (Reach × Impact × Confidence) / Effort`

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Reach = number of assets or users affected per month. Divide by an arbitrary scale factor (10 for assets, 1 for users) to keep numbers readable.

Priority Ranking (High → Low)

1. **SSE Real-Time Progress** (168.0) — Quick win; high user delight 2. **Violation Viewer** (153.0) — Core UX differentiator 3. **AI Scan Engine** (90.0) — Platform backbone (must-build despite lower RICE due to dependencies) 4. **S3 Pre-Signed Upload** (285.0) — Trivial effort, high value 5. **Email Notifications** (94.5) — Critical for workflow completion 6. **Batch Scanning** (57.6) — Post-MVP; high effort needs validation 7. **Role-Based Workflow** (33.3) — Required for compliance process but lower per-scan frequency 8. **Dashboard & Analytics** (14.0) — Phase 2 9. **Immutable Audit Trail** (5.5) — Phase 2; small audience 10. **Rule Engine Admin UI** (3.5) — Phase 3; administrative overhead

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Success Metrics

KPI	Baseline	Target (3 months)	Target (6 months)	Measurement Tool
Average compliance cycle time	3.8 days	≤ 1 day	≤ 4 hours	App analytics
Violation detection rate	92%	97%	99.5%	Manual audit of 10% sample
User adoption (weekly active users)	N/A	20 users	30+ users	Auth0 analytics
False positive rate	N/A	≤ 15%	≤ 8%	Checker rejection analysis
Checker throughput (assets reviewed/hour)	2 assets/hr	8 assets/hr	15 assets/hr	System metrics
Penalty exposure reduction	₹25L max	₹10L max	₹0	External audit
NPS (internal user survey)	N/A	≥ 40	≥ 60	Quarterly survey

Roadmap

Phase 1 — MVP (Weeks 1–8)

Week	Deliverable
1–2	Next.js 16 project setup, FastAPI scaffold, S3 setup, Upstash Redis
3–4	AI Scan Engine: Tesseract OCR pipeline + 6 core IRDAI rules
5	Violation Viewer with bounding boxes (React Canvas overlay)
6	SSE real-time scan progress; asset upload with pre-signed URLs
7–8	Maker/Checker workflow + Resend email notifications; UAT with 5 users

Phase 1 — MVP (Weeks 1–8)

Go/No-Go criteria: 95% OCR accuracy on test assets; cycle time ≤ 1 day; false positive rate ≤ 20%

Phase 2 — Growth (Weeks 9–16)

Go/No-Go criteria: All 12 rules operational; audit trail verified; load test passes at 100 concurrent users

Phase 3 — Scale (Weeks 17–24)

Impact Estimate

Quantitative Impact (Annualized, Post Phase 2)

Metric	Current	Projected	Savings / Value
Compliance cycle time	3.8 days	4 hours	89% faster launches
Compliance team FTEs needed	4 FTEs	1.5 FTEs	₹21 Lakh/year savings
Penalty exposure	₹25 Lakh/violation	₹0 (target)	₹75 Lakh–₹3.75 Cr risk eliminated
Marketing opportunity cost (delayed launches)	₹2.1 Cr/year	₹25 Lakh	₹1.85 Cr recovered
Audit preparation time	2 weeks/quarter	2 hours/quarter	95% reduction

Quantitative Impact (Annualized, Post Phase 2)

Total Estimated Annual Value: **₹3.5–₹5.0 Cr**

Qualitative Impact

- **Compliance team morale**: Reduced burnout from manual review
- **Regulatory relationship**: Demonstrable compliance rigor to IRDAI
- **Brand safety**: Zero non-compliant assets reaching market
- **Time-to-market**: Campaign launches aligned with optimal windows

Break-Even Analysis

- Total development cost (6 months, 4-person team): **₹48 Lakh**
- Annual infrastructure (AWS S3 + Upstash + Resend + FastAPI server): **₹3.6 Lakh**
- Annual maintenance (20% of dev cost): **₹9.6 Lakh**
- **Break-even: 4 months** post-launch (based on FTE savings alone)
- **ROI: 8.2× in Year 1**

Document version: 1.0 | Next review: 2026-06-12